

Junhyuk Heo

✉ E-mail  Github  LinkedIn  Google Scholar  Homepage

Introduction

My work focuses on machine learning for Earth observation, aiming to derive real-world value from satellite imagery for environmental monitoring and climate change. My interests include learning methods for extracting meaningful information from remote sensing data (e.g., target detection and semantic segmentation), as well as representation learning and foundation models. I also work on satellite data preprocessing, including atmospheric and radiometric correction, and I am increasingly interested in onboard processing and compression for more efficient use of satellite observations.

Education

Konkuk University
Major in Civil Engineering
Minor in Computer Science

Mar 2018 – Feb 2024
Seoul, Republic of Korea

Publications

- Few-Shot Open-Vocabulary Remote Sensing Segmentation via Textual Inversion** *ECCVw (Oral), 2026*
Junhyuk Heo, Junghwan Park
[Paper](#) [Code](#)
- FLAME: Physics-Guided Neural Operators for Onboard Satellite Methane Detection in Hyperspectral Imagery** *ICMLw, 2026*
Junhyuk Heo, Junghwan Park, Sangcheol Sim, Beomkyu Choi, and Woojin Cho
[Paper](#) [Code](#)
- MANGO: A Global Single-Date Paired Dataset For Mangrove Segmentation** *IGARSS (Oral), 2026*
Junhyuk Heo, Beomkyu Choi, Hyunjin Shin, and Darongsae Kwon
[Paper](#) [Code](#)
- ELMZip: Onboard Satellite Image Compression via Extreme Learning Machines for Efficient Downlink** *IGARSS, 2026*
Woojin Cho, Junghwan Park, Sangcheol Sim, Steve Andreas Immanuel, *Junhyuk Heo*, and Darongsae Kwon
[Paper](#)
- Basis-Oriented Low-rank Transfer for Few-Shot and Test-Time Adaptation** *CVPR, 2026*
Junghwan Park, Woojin Cho, *Junhyuk Heo*, and Darongsae Kwon
[Paper](#)
- Self-Supervised Score-Based Despeckling for SAR Imagery via Log-Domain Transformation** *Preprint, 2026*
Junhyuk Heo
[Paper](#) [Code](#)
- Fourier-Modulated Implicit Neural Representation for Multispectral Satellite Image Compression** *IGARSS (Oral), 2025*
Woojin Cho*, Steve Andreas Immanuel*, *Junhyuk Heo*, and Darongsae Kwon
[Paper](#) [Code](#) [Poster](#)
- Tackling Few-Shot Segmentation in Remote Sensing via Inpainting Diffusion Model** *ICLRw (Best Paper Award), 2025*
Steve Andreas Immanuel, Woojin Cho, *Junhyuk Heo*, and Darongsae Kwon
[Paper](#) [Code](#) [Poster](#)
- SAR-to-optical image translation with UNSB-FFC considering shadow area pixels** *KSCE Journal of Civil Engineering, 2025*
Su Min Jo, *Junhyuk Heo*, and Yang Dam Eo
[Paper](#)
- Neural Compression for Multispectral Satellite Images** *NeurIPS w, 2024*
Woojin Cho*, Steve Andreas Immanuel*, *Junhyuk Heo*, and Darongsae Kwon
[Paper](#) [Code](#) [Poster](#)

Research Experiences

TelePIX [🔗](#)

AI Researcher

May 2024 – Aug 2026

Republic of Korea

- **Satellite image preprocessing:** Developed workflows for inter-band registration, radiometric and atmospheric correction, geometric correction, and orthorectification.
- **Onboard image processing:** Implemented and tested model inference on NVIDIA Jetson using TensorRT for onboard satellite image processing.
- **Segmentation under limited supervision:** Researched semantic, few-shot, and open-vocabulary segmentation across optical and SAR imagery under limited annotations and diverse environmental conditions.
- **Satellite image compression:** Researched multispectral image compression using modulated implicit neural representations and contributed to lightweight compression methods for efficient satellite downlink.
- **Earth foundation models:** Explored pretrained models and geospatial embeddings for satellite scene representation and downstream analysis.

Geomatics Lab [🔗](#)

Research Intern

Jun 2023 – Feb 2024

Konkuk Univ., Republic of Korea

- Analyzed satellite imagery using QGIS and ArcGIS.
- Researched SAR-to-optical image translation using generative models.

Projects

EarthScope: Interactive Earth Observation Analysis [🔗](#)

Python, FastAPI, PyTorch, Google Earth Engine

Aug 2025 – Jun 2026

[Demo](#) [🔗](#)

- Built a web-based platform for Sentinel-1/2 and user-supplied GeoTIFF imagery, integrating spectral indices, segmentation, change detection, and geospatial visualization.
- Combined classical spectral/statistical detection (Spectral Angle Mapper, ACE, CEM, RXD) with deep learning segmentation (SAM 3, SegFormer, U-Net++), with GPU-accelerated analysis and interactive mask comparison.

Global Mangrove Monitoring [🔗](#)

TelePIX, 2026

- Developed deep learning models to segment mangroves in Sentinel-2 imagery for coastal blue-carbon monitoring.
- Compared models with spectral-index baselines and evaluated generalization using country-disjoint MANGO splits.
- Project selected for the **Google DeepMind Accelerator: AI for the Planet in APAC (2026)**.

Training-Free Scale Adaptation of LVLMs for Remote Sensing Imagery [🔗](#)

Dec 2025

AGU Fall Meeting – Poster Abstract

- Investigated scale mismatch in remote sensing imagery and inference-time adaptation of frozen large vision-language models without fine-tuning model parameters.
- Explored multi-scale visual representations and scale-adaptive token weighting; accepted as an AGU25 poster abstract.

Hyperspectral Image Pansharpening (1st author) [🔗](#)

Jan 2025

ESA-NASA Workshop on AI Foundation Models for Earth Observation – Poster Abstract [🔗](#)

- Proposed a panchromatic-conditioned INR for continuous-resolution pansharpening, addressing resolution mismatch and spectral distortion in hyperspectral imagery.
- Designed for single-forward-pass inference without per-image optimization; proposal accepted for presentation at the ESA-NASA workshop.

Deep Learning Projects [🔗](#)

Jul 2022 – Jan 2024

deep daiv. – Deep Learning Club [🔗](#)

- Developed a golf swing analysis web service using YOLO-based 3D pose estimation and a model-to-web pipeline for change detection.
- Built projects in text-based meme recommendation, face synthesis, and object-to-Hanjii image stylization.

Research Interests

Segmentation & Target Detection: Classical spectral/statistical methods, deep learning, and few-shot learning.

Earth Observation Foundation Models: Geospatial representations, multi-sensor learning, and data-efficient adaptation.

Vision Foundation Models: Vision-language models, promptable and open-vocabulary segmentation.

Implicit Neural Representations: Satellite image compression, continuous image representations, and image reconstruction.

Skills

Programming Languages: Python, C/C++, JavaScript, MATLAB

Deep Learning: PyTorch, TensorFlow, Keras, Distributed Data Parallel (DDP), TensorRT

Remote Sensing: ArcGIS, QGIS, Google Earth Engine

Tools & Platforms: Git, Docker, Linux, AWS, NVIDIA Jetson, FastAPI, L^AT_EX

Languages: Korean (native), English (advanced; OPIc AL)